

ChE 382: Unit Operations Laboratory

BINARY BATCH DISTILLATION

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Introduction

Distillation is the most widely used separation process in chemical engineering. It can handle a wide variety of feeds and has the potential to produce very high purity products. At times, a petrochemical plant appears to consist mainly of distillation columns involved in product purification. When a chemical engineer faces a separation problem in process design, the usual first alternative is distillation.

Distillation is classified as an equilibrium separation process since the gas and liquid phases are expected to reach phase equilibrium on each contacting stage of the distillation column. (The other major class of separations is rate-governed processes such as filtration, which depend upon the relative rates of flow of the components under a driving force).

Equilibrium on each stage is a theoretical concept which cannot be totally achieved in practice. Chemical engineers have developed models for distillation which take into account this non-ideal behavior. Thus, the concept of efficiency is introduced into the analysis of the separation process.

In this experiment, you will examine a **Binary (two component) Distillation Column** for separating ethanol and water. In distillation, the two phases are liquid and vapor. They are separated simply by gravity; the liquid falls, and the vapor rises. Thus, the two phases travel **countercurrently** through the column, which always results in a greater separation than **cocurrent** processes. However, countercurrent systems require trial and error calculations for a complete solution. The **McCabe-Thiele** method provides an approximate, graphical method for solving all countercurrent equilibrium staged separation problems.

A good review of distillation technology and theory can be found in the references [1] – [5].

Basic Theory

The basic process concept for a continuous distillation column is shown below. The feed enters near the middle of the column. Heat is supplied in the reboiler to provide the energy to boil the mixtures. The product vapors are condensed at the top of the column. Within the column are the stages where gas and liquid contact; the liquid flowing down the column, the vapor up the column. Ideally, there is vapor-liquid equilibrium on each stage of the column

An important process variable is the reflux ratio at the top of the column. This plays an essential role in controlling the product purity and in determining the energy which is

required in the reboiler. Also of importance is the condition of the feed stream (e.g., saturated liquid) entering the column.

Batch distillations are also used commercially and are easier to analyze and control. In these processes, the feed is placed in the still at the beginning of the run and product is taken off the condenser as a function of time. It is an unsteady-state process.

A complete simulation of the distillation column would include the following balances:

1. Overall mass balance for the column. A mass balance for each component and the overall balance.
2. Mass balance for each of the equilibrium stages.
3. An overall heat balance for the column.
4. A heat balance for each stage. This may need to take into account the heat of mixing of the components.

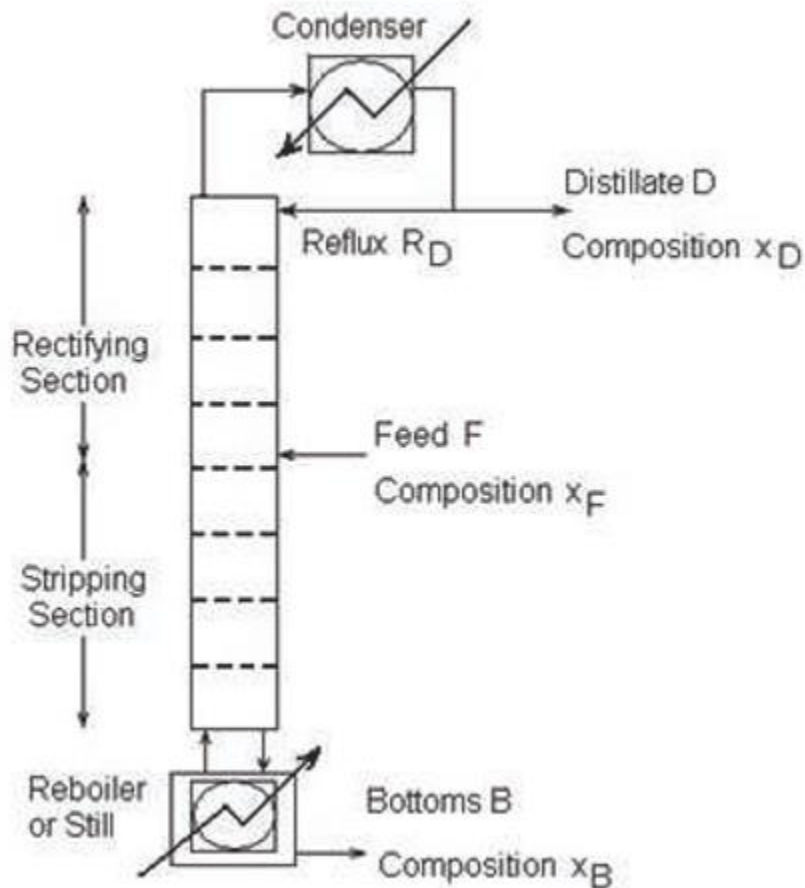


Figure 1. Schematic of a distillation column.

This can become a very complex set of equations for a multi-component distillation column. There are numerous sophisticated computer programs available to analyze these columns to let the chemical engineer devise an operational and efficient design. Examples of design packages widely used are CHEMCad, HYSYS, and Aspen. Many companies have their own proprietary design tools.

McCabe-Thiele Analysis for Binary Distillation

For a binary system, the classic McCabe-Thiele graphical technique can be used to design the distillation column and to explore the effect of process variables. Although it is quite simple to use, this simplicity comes with a price. The technique uses a number of assumptions concerning the operation of the distillation column. It is important to analyze whether these assumptions are reasonable for the batch distillation unit being used.

To use the McCabe-Thiele analysis, one needs the vapor-liquid equilibrium for the system plotted as a y versus x diagram. The second part is the operating line for the column which is a formulation of the mass balance from stage to stage; this requires the overall gas and liquid flows and the selected reflux ratio. When the distillation is continuous and feed enters the middle of the column, there are separate operating lines (mass balances) for the column above and below the point of feed introduction. For batch distillation, there is only one operating line since the molar flows of gas and liquid are constant throughout the column.

One may examine the assumptions involved in deriving this operation line. The number of stages is determined by moving from equilibrium to operating curve such as occurs in the column. This generates a step structure which gives the number of stages. While the method is simple to do in practice, it takes some analysis to understand the concepts.

A typical McCabe-Thiele diagram is shown below:

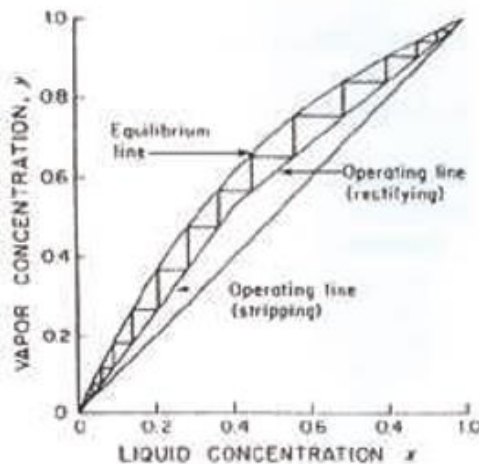


Figure 2. McCabe-Thiele diagram.

Column Efficiency

The basic McCabe-Thiele method assumes vapor-liquid equilibrium on each stage. Many chemical engineers have invested major efforts in devising mechanical devices to promote the attainment of equilibrium on the stage. However, this equilibrium is never obtained completely in practice and in many situations the assumption of equilibrium is a poor one.

The efficiency is introduced with the concept of Murphree plate efficiency, which is defined as:

$$\eta = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad (1)$$

Where y_n^* is the composition in equilibrium with x_n . If the vapor and liquid are in thermodynamic equilibrium, y_n equals y_n^* . The figure below shows the practical effect on the McCabe-Thiele diagram; inclusion of the plate efficiencies lowers the equilibrium curve to the dashed line.

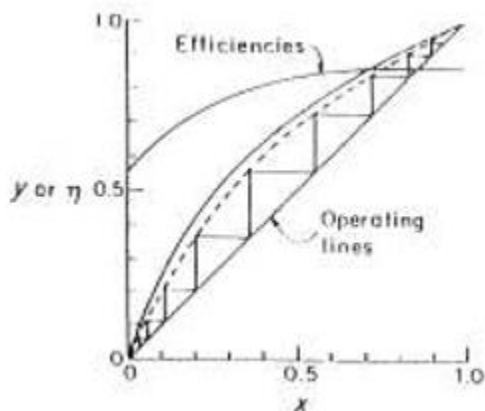


Figure 3. McCabe-Thiele analysis

Armfield Distillation System

The Armfield UOP3CC Continuous Distillation Column is a self-contained distillation facility consisting of two interconnected units; a floor-standing process unit and a bench-mounted control console. A PC computer is also connected to the console. The column can be operated from the console or the computer using Armfield software. **NOTE: Before running the experiments, read over the Armfield Instructional Manual [6] to become familiar with the equipment, operation procedures and safety guidelines.**

Experimental Procedure: Overview

A 10 wt% ethanol solution will be used for the 2 sets of experiments below:

Experiment A: Effect of Boil-Up Rate and Overall Column Efficiency

Experiment B: Distillation at Constant Reflux Ratio.

200-proof ethanol is available in 4L containers in the fume hood on the south wall of the lab. If there is not enough ethanol in the container, let the TA or instructor know so they can provide a new bottle. Read the MSDS for ethanol to understand the hazards and handling procedures; denatured ethanol contains solvents and is not safe for consumption. Fill the reboiler with the ethanol solution to 10 liters. Do not turn on the distillation column until you check with TA or instructor. You can monitor the progress of the distillation column by watching the tray temperatures as a function of time using the control console or the software. Each tray will warm up as liquid accumulates on the tray. These temperatures can be used to estimate the composition of the liquid on the tray assuming that the trays are at equilibrium. Use these temperatures to examine the operation of the column, and to monitor its approach to steady-state operation. Vapor-liquid equilibrium for ethanol-water is available in the literature.

Experiment A: Effect of Reboiler Power at Total Reflux

Objective

To determine the variation of boil-up rate, pressure drop over the distillation column, and overall column efficiency as a function of reboiler power at total reflux.

Method

While operating at total reflux, the boil-up rate and pressure drop over the distillation column is measured for different reboiler power rates. In addition, overhead and bottom liquid samples are taken across the range of boil-up rates to determine the ethanol composition and to calculate the overall column efficiency.

Theory

The total pressure drop across each tray is the sum of that caused by the restriction of the holes in the sieve tray, and that caused by passing through the liquid (foam) on top of the tray.

As the velocity of the vapors passing up the column increases the overall pressure drop also increases. The velocity is controlled by varying the boil-up rate which is done by varying the power input to the reboiler. Under conditions with no liquid present, the sieve trays will behave like an orifice in that the pressure drop will be proportional to the square of the velocity. Since there is a liquid head however, this square relationship does not become apparent until the head of liquid has been overcome and foaming is taking place. In

a graph of pressure drop vs. boil-up rate (log/log), at low boil-up rates the pressure drop will remain fairly constant until foaming occurs when the pressure drop would be expected to rise sharply for unit increases in boil-up rate.

To calculate the number of theoretical plates for a given separation at total reflux, Fenske developed the following formula:

$$n + 1 = \frac{\log \left[\left(\frac{x_{EtOH}}{x_{H2O}} \right)_d \left(\frac{x_{H2O}}{x_{EtOH}} \right)_b \right]}{\log(\alpha_{EtOH/H2O})_{avg}}$$

where n = number of theoretical plates

x_{EtOH} = mole fraction of more volatile component (ethanol)

x_{H2O} = mole fraction of least volatile component (water)

Average relative volatility: $(\alpha_{EtOH/H2O})_{avg} = \sqrt{\alpha_d \alpha_b}$

Subscripts d, b indicate distillate and bottom respectively. The efficiency is given by;

$$E = \frac{\text{Number of theoretical plates}}{\text{Number of actual plates}} \times 100\%$$

Knowing the composition of distillate and bottom and the corresponding volatilities, the column efficiency can be determined.

Equipment Set Up

Make up 10 liters of a mixture of 10 wt% ethanol/water. Set the equipment to operate at total reflux by switching off the reflux ratio timer on the console. Refer to Armfield Instruction Manual for a diagram of the process unit and location of the unit components (valves, etc.)

Ensure all valves on the equipment are closed.

Open valve V10 on the reflux pipe.

Load the 10 liter of ethanol/water mixture through the filler cap on top of the reboiler using a funnel. Make sure the filler cap on the top of the reboiler is firmly replaced after filling.

Procedure

Turn on the power to the control panel.

Set the temperature selector switch to T9 (the temperature in the reboiler).

Open valve V5 until the cooling water flow rate FI1 to the condenser is approximately 3 litres/min.

On the control panel, turn the power controller for the reboiler heating element to 1.50 kW and switch the switch turning on the power to the heating element to “power on” position. Another red lamp will illuminate indicating the heating element is on. Open valves V6 and V7 which connect base and top of the distillation column, respectively, to the manometer. Initially there will be no pressure difference in the column meaning no reading on the manometer. Close valve V6 and V7.

The 1st and 5th plates in the column sections are not insulated so that observation of the sieve plates is possible. Make notes on the appearance of these trays throughout the distillation process.

Eventually, vapor will begin to rise up the column and the progress of this can be clearly observed as well as detected by the increasing temperatures when switching the temperature selector on T8, T7, T6, T5, T4, T3, T2 and T1. Vapor will enter the condenser and reappear as droplets into the glass walled distillate receiver vessel (11). The distillate will build up a small level in the receiver and eventually overflow to the reflux regulator valve (12). Since the valve is not on (on the control panel), and it will not be necessary to have it on in this experiment, which is run under total reflux, the condensed vapor will return to the column.

After some time total reflux will occur (steady bubbling on all trays). At this point turn the power controller clockwise until a reading of approximately 0.75kW is obtained on the digital wattmeter. The cool distillate is then returning on the top of the column, and will cascade down the trays forming a liquid level on the trays and bubbling of vapor passing through the liquid. The system will have reached an equilibrium condition at the given power rate when the temperatures T1, T2, T3, T4, T5, T6, T7 and T8 are constant.

Measure the boil-up rate at 0.75kW by performing a timed volume collection: Operate valve V3 so that all the condensate is diverted into a measuring cylinder, and use the stopwatch to measure the time required to collect a set quantity. This will not disrupt the equilibrium conditions in the column provided a liquid level is maintained in the condensate feeding pipe. When taking a sample, partially open valve V3 and drain the condensate (in a separate measuring cylinder) from the reflux system until a steady flow is obtained. (Ensure that liquid remains in the flexible connecting tube to prevent vapor from escaping.) Start sample collection and timing at the same time, and collect a sizeable amount. Pour the first non-representative collected amount in a bottle to be discarded later.

After taking the samples, take readings of pressure drops over both the rectifying (top) and the stripping (bottom) sections by opening the valves V6 and V7 on the manometer. When opening the valves, make sure always to open valve V6 then V7 to prevent vapor from the column entering the manometer. If this happens, it can be seen as two separate phases in the U-tube. Close the valves in the same order after taking the pressure drop reading.

After the representative sample has been taken, keep the sample tubes in an upright position. Do not overturn them because of the possibility of evaporation of the sample. Measure the composition of the samples (% v/v) using the Snap 41 Densitometer (Note: we are not using the Refractometer). Follow the measurement procedure in the Snap 41 manual. If needed, ask the instructor for help.

Following a similar procedure, take a sample of the bottom through valve V2.

CAUTION! THIS SAMPLE WILL BE HOT!

Record the composition for this sample, too using the Snap 41 densitometer.

Repeat this procedure until at least 3 samples of both overhead and bottom are obtained. Record the temperatures T8 and T1 to calculate the average column temperature.

Step up the boil-up rate in 250 Watt increments up to maximum 1.50 kW by adjusting the boiler heater power controller (on the control panel). Take similar readings of the boil-up rate, pressure drops, overhead and bottom composition after having allowed the column to stabilize.

Boil-Up Rate Results

Tabulate your results under the following headings:

Power (kW)	Boil-up Rate (litres/hr)	Pressure Drop (cm H ₂ O)			Degree of Foaming On Trays
		Top	Bottom	Overall	
0.75					
1.00					
1.25					
1.50					

From Fenske's equation: $n = 5.2$, i.e. the number of theoretical plates is 5.2.

Similarly for each sample, the column efficiency, E , can be calculated

Additional Data Analysis

1. Plot the temperature recorded as a function of time.
2. Briefly explain these temperatures. How does the column approach steady state?
3. Comment on the relationship observed, if any, between boil-up rate and pressure drop.
4. Relate the descriptive comments observed to zones on the curve.
5. Is there any obvious relationship between boil-up rate and column efficiency?
6. How does the efficiency of the column compare to full-scale columns in use in industry?

Experiment B: Distillation at Constant Reflux Ratio.

Objective: To carry out a distillation at constant reflux ratios, thereby varying top and bottom compositions with time.

Theory

At total reflux, since there is no feed, no bottom product or no top product, the liquid flow in the column is equal to the vapor flow in the column:

$$V = L$$

A material balance over the M.V.C. (most volatile component) gives

$$V y_n = L x_{n+1}$$

Since $V = L$ this gives

$$y_n = x_{n+1}$$

Experimental Procedure

Use the ethanol/water mixture that is already in the column from the previous Experiment A and which is already at a high temperature. Set the heat controller to give steady bubbling on all tray and total reflux, about 0.75kW. Leave the apparatus for at least 15 minutes so that the system can reach an equilibrium condition. Record the temperatures.

This experiment is run at varying reflux ratios. Set the reflux controller to the desired values e.g. start with 5:1 by setting 20 sec back to column and 4 sec to top product receiver. The setting is like the setting of a digital watch. C- means back to column and C+ means to

top product receiver. When this is done, switch the reflux valve on (on the control panel). You should also see condensed vapor flowing to the top product receiver.

Measure the boil-up rate by performing a timed volume collection: Operate valve V3 so that all the condensate is diverted into a measuring cylinder. First take a small discarding sample of 5 to 10 ml into an alternative vessel, then divert the condensate into the measuring cylinder and use the stopwatch to measure the time required to collect a set quantity. This will not disrupt the equilibrium conditions in the column provided a liquid level is maintained in the condensate feeding pipe. When taking a sample, partially open valve V3 and drain the condensate (in a separate measuring cylinder) from the reflux system until a steady flow is obtained. (Ensure that liquid remains in the flexible connecting tube to prevent vapor from escaping.) Start sample collection and timing at the same time, and collect a sizeable amount. Repeat the measurement three times and take an average value to determine the boil-up rate.

Take a sample of the overheads through valve V3. When doing so be careful never to drain the condensate return line, i.e. partially open valve V3 to leave a small amount of liquid in the line all the time.

Generally, when taking samples, drain a “discarding” sample of approximately 5 to 10 ml before taking the representative sample in a sample glass. Do not drain too much of the “discarding” sample because of the disturbance of the mass balance. Discard the “discarding” sample in a safe way. After the representative sample has been taken, keep the sample glasses in an upright position. Do not overturn them because of the possibility of evaporation of the sample.

Record the % v/v for the taken overhead sample. In a similar manner **and preferably at the same time** take a sample of the bottom through valve V2 and measure the % v/v.

CAUTION! THIS SAMPLE IS HOT!

Repeat this procedure every 5 minutes until at least three samples of both overhead and bottom are obtained. Record the temperatures T8 and T1 to calculate the average column temperature.

Repeat this procedure for at least one different reflux ratio.

Results

SAMPLE CALCULATION (methylcyclohexane/toluene mixture)

Top product required: 75 mol percent methyl*cyclohexane*

Bottom product required: 37 mol percent methyl*cyclohexane*

From the equilibrium curve,

Composition on the top sieve plate is $y_t = 0.78$, $x_t = 0.75$

and

$$y_{t-1} = x_t = 0.75$$

$$x_{t-1} = 0.72$$

$$y_{t-2} = x_{t-1} = 0.72$$

$$x_{t-2} = 0.69$$

$$y_{t-3} = x_{t-2} = 0.69$$

$$x_{t-3} = 0.65$$

$$y_{t-4} = x_{t-3} = 0.65$$

$$x_{t-4} = 0.60$$

$$y_{t-5} = x_{t-4} = 0.60$$

$$x_{t-5} = 0.54$$

$$y_{t-6} = x_{t-5} = 0.54$$

$$x_{t-6} = 0.47$$

$$y_{t-7} = x_{t-6} = 0.47$$

$$x_{t-7} = 0.40$$

$$y_{t-8} = x_{t-7} = 0.40$$

$$x_{t-8} = 0.37$$

This composition is close to that required and can be withdrawn from the bottom product.

Theoretically, therefore, the distillation column containing eight sieve plates plus the boiler will give the compositions calculated above. However, as the experiment will show this is not in fact correct. Relate the theory and the results of the experiment with the previous experiment to determine column efficiency.

The previous calculation can be illustrated in a so-called McCabe-Thiele diagram:

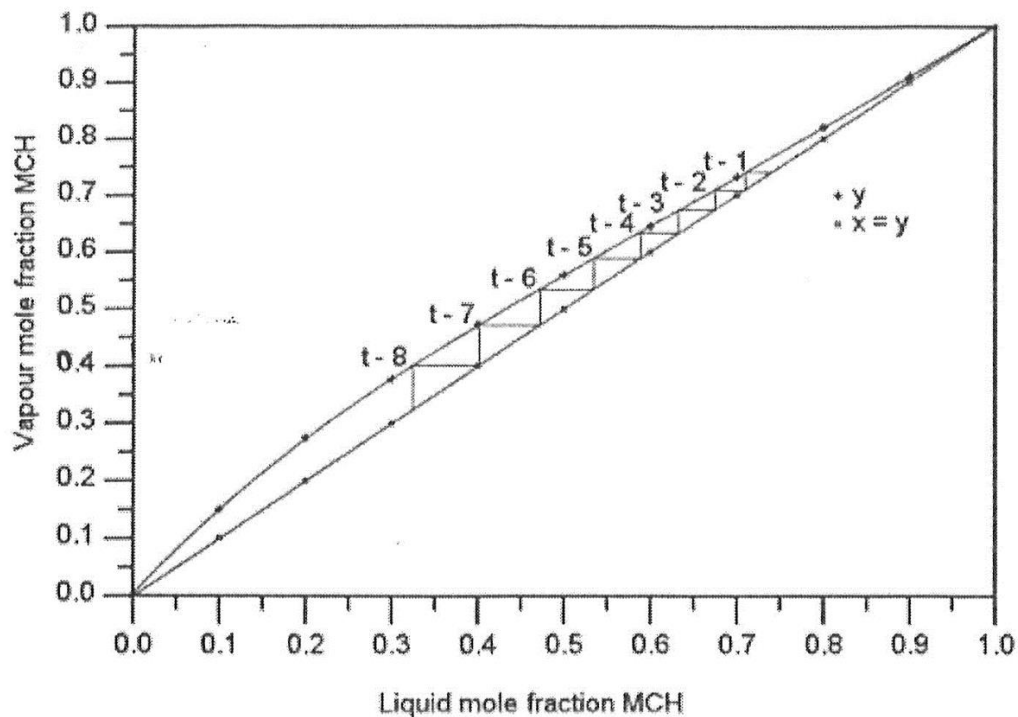


Figure 4. McCabe-Thiele analysis for Methylcyclohexane/toluene.

Total condensate flow rate

- a)
- b)
- c)
- d)
- e)

Average flow rate =

Top Product Composition	Bottom Product Composition
Mol Percent methylcyclohexane	Mol Percent methylcyclohexane

Additional Data Analysis

Refer back to the Theory section of this experiment. Did the results obtained in this experiment correlate with the McCabe-Thiele diagram presented in the theory section?

Explain the differences between your results and the ideal McCabe-Thiele analysis. Which assumptions are valid, and which are invalid? Can you explain your results by assuming that the trays are not 100% efficient? Can you explain your results by assuming that energy is lost in the column? To do this, perform an overall energy balance on the column using the reboiler and condenser heats. Assume that the energy carried away in the two product streams balances the energy added in the feed stream

References

- [1] J. Kunesh, H. Kister, and et al., "Distillation: Still towering over other options," *Chemical Engineering Progress*, vol. 89, p. 43, 1995.
- [2] C. J. King, *Separation Processes*, 2nd ed. McGraw-Hill, 1980.
- [3] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, 7th ed. McGraw-Hill, 2005.
- [4] R. H. Perry and D. W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7th ed. McGraw-Hill, 1997.
- [5] P. C. Wankat, *Equilibrium-Staged Separations*. Prentice Hall, 1988.
- [6] Armfield Limited, UOP3CC Continuous Distillation Column Instruction Manual, Issue 38 March 2014.

Figure 5. Schematic of Armfield Distillation Column

